Nikhil Raushan

+91 8789120181 — nikhilraushan470@gmail.com — LinkedIn — GitHub

Technical Skills

Domains: Data Structures and Algorithms, Problem Solving, Full-Stack Web Development
Languages: C, C++, HTML, JavaScript, CSS, SQL
Frameworks & Tools: React.js, Express.js, Node.js, MongoDB, Tailwind CSS, Cloudinary
CS Coursework: DSA, Object-Oriented Programming, DBMS, Computer Networks
Databases and Caching: MongoDB, MySQL

Projects

Online Code Compiler with Integrated AI (MERN Stack)
React.js, Node.js, Express.js, Monaco Editor, OpenAI API

- Developed a full-stack online code compiler using React and Node.js following a client-server architecture.
- Integrated Monaco Editor to provide a VS Code-like coding experience with syntax highlighting and multi-language support.
- Added AI-powered code suggestions and code review functionality using OpenAI API.
- Implemented compilation and runtime error handling using stdout, stderr, and execution logs.

BidDaddy – Auction Platform (Full Stack)
React.js, Node.js, Express.js, MongoDB, JWT, Cloudinary

- Developed a full-stack auction platform where users can register, log in, create auctions, and place bids.
- Implemented secure authentication and authorization using JWT and bcrypt password hashing.
- Integrated Cloudinary for image upload and cloud-based media storage.
- Developed responsive frontend components using React.js and React Router for seamless navigation.

Smart Home Energy Optimization System
React.js, Node.js, Express.js, MongoDB, Python, Flask, Scikit-learn

- Built a full-stack smart home energy monitoring system using the MERN stack with machine learning integration.
- Integrated a Python Flask-based ML service to predict future energy usage using Random Forest regression.
- Implemented REST APIs to collect, store, and manage household energy consumption data using MongoDB.

SIW Antenna Design and Fabrication

- Designed and simulated a Ka-band SIW antenna using ANSYS HFSS for high-frequency communication applications.
- Analyzed antenna performance using S-parameters, impedance matching, and radiation characteristics.
- Evaluated antenna characteristics including reflection coefficient (S11) and bandwidth

Education

BMS College of Engineering, Bengaluru
B.E. in Electronics and Communication Engineering — 2022 – 2026 (Expected)

Achievements

- Solved 250+ DSA problems across LeetCode, Code360, and GeeksforGeeks.

Extracurricular Activities

- Event Volunteer – College Fest 2023: Assisted in managing main stage events and stalls.
- Design Volunteer – BMSCE Incubation Cell (BIG Foundation): Worked on design and branding tasks.